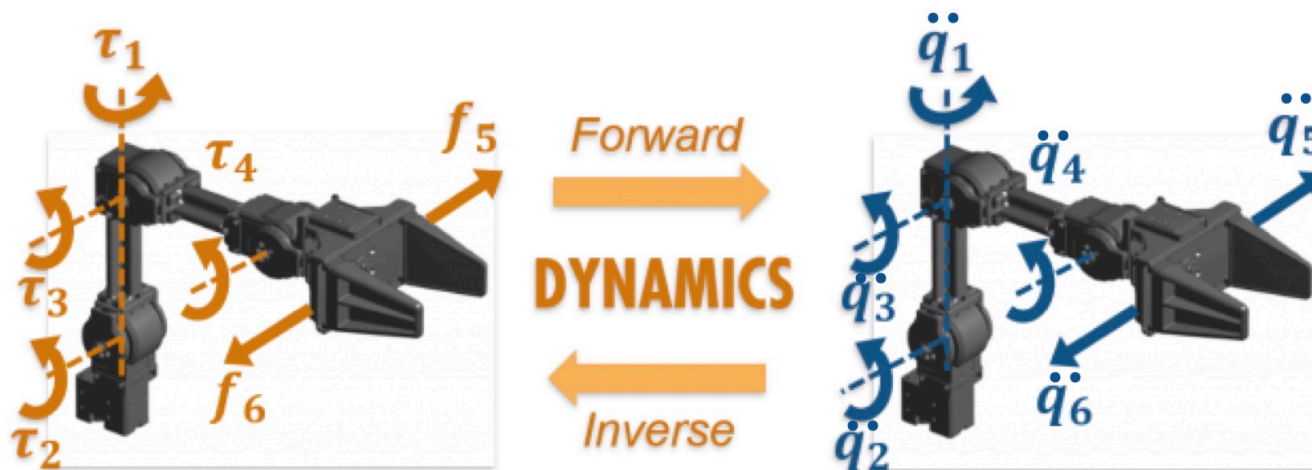


Robot Learning

Lecture 2b. Dynamics of Robot Manipulators

Dynamics

- Dynamics studies the relationship between the **forces** acting on a robot mechanism and the resulting **accelerations and motion trajectories**



Dynamics

- Dynamics are important to account for the **physical interactions** of the robot with the **environment**



Dynamics

- Dynamics are important to account for the **physical interactions** of the robot with the **environment**

Example: opening a door



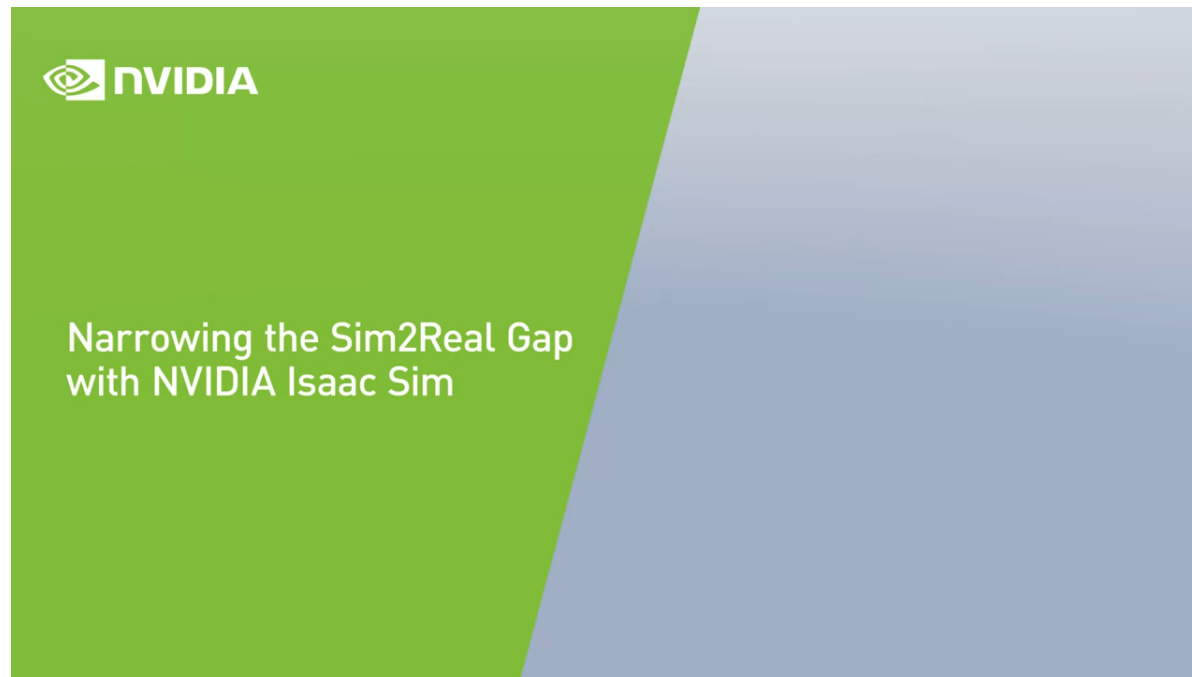
A.Y. 2025-2026



Robot Learning

Dynamics

Having a good representation of the dynamic behavior of robots and their interactions with the environment is also critical to build **simulations** that allow for a more reliable development of algorithms and facilitate their transfer to the real world.



Canonical equations

The equations of motions of a robot are typically represented in one of two forms, called **canonical equations**

$$\Lambda(\mathbf{x})\dot{\mathbf{v}} + \boldsymbol{\mu}(\mathbf{x}, \mathbf{v}) + \boldsymbol{\rho}(\mathbf{x}) = \mathbf{f}$$

Task-space formulation

$$B(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}$$

Joint-space formulation

Generalized Inertia matrix:
symmetric, $n \times n$, positive-
definite matrix

$n \times n$ matrix with the
Coriolis and
Centrifugal terms

vector of **gravity**
terms

vector of
generalized forces

Approaches to dynamic modeling

$$B(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}$$

The terms in this canonical equation can be derived using different methods

Lagrange method

- Energy based approach
- Dynamic equations obtained in closed form

Newton-Euler method

- Approach based on the balance of forces/torques
- Dynamic equations recursively (numeric method)

Lagrange formulation

The Lagrange formulation proceeds via the **Lagrangian** of the robot mechanism

$$L = T - U$$

Total **kinetic** energy

Total **potential** energy

The Lagrange equation is

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \tau_i$$

Generalized coordinates

Non-conservative
Generalized forces

Generalized coordinates and forces

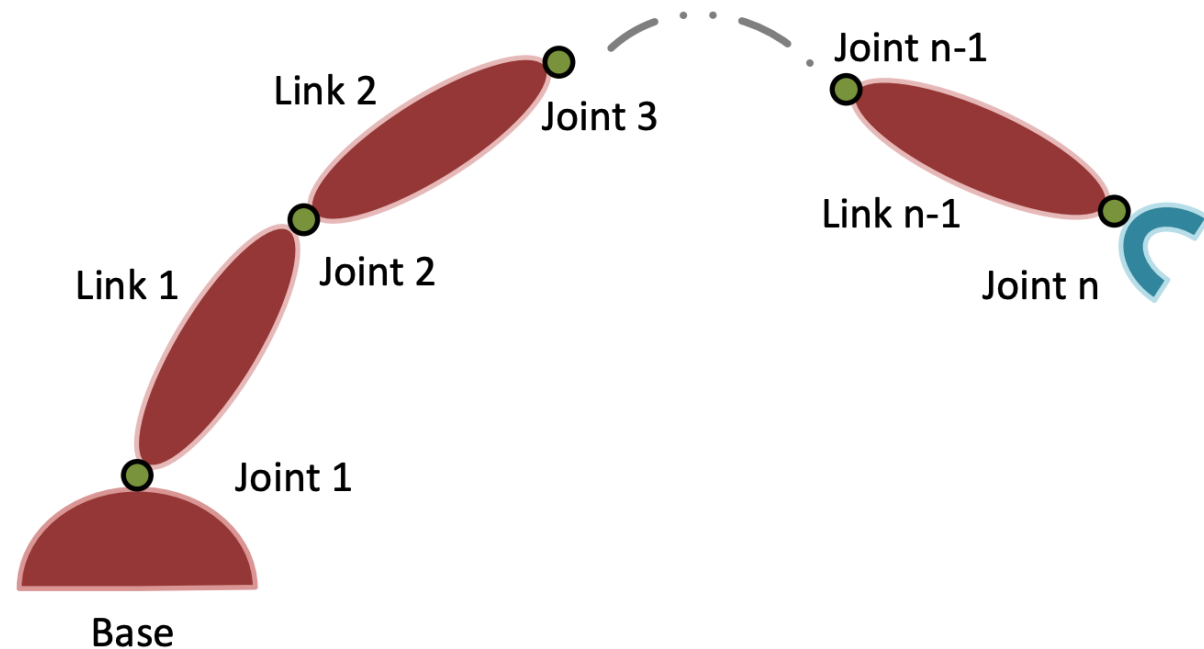
- The equations of motion can be formalized in a number of **different coordinate systems**
- **n independent coordinates** are necessary to describe the motion of a system having **n degrees of freedom**
- any **set of independent coordinates** is called generalized coordinates: $q_1, q_2, \dots, q_n$
- these coordinates may be **lengths, angles**, etc. For a manipulator, we can use the joint variables.
- The **generalized forces** are the forces that act on the generalized coordinates.

Kinematic chain

- A serial manipulator can be regarded as a chain of rigid bodies
- We can express the **total kinetic and potential energy** of the robot as the **sum of the contributions of the individual links**

$$T = \sum_{i=1}^n T_i$$

$$U = \sum_{i=1}^n U_i$$



Kinetic energy of a link

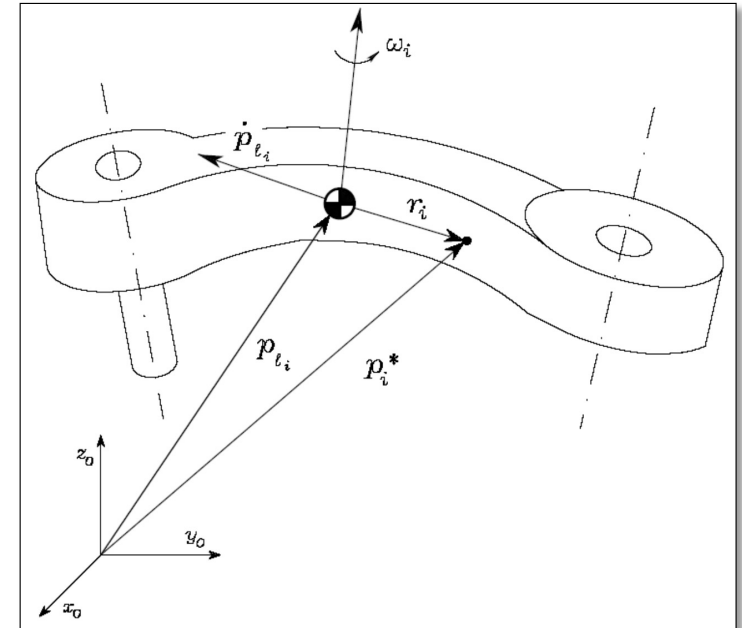
Consider a single rigid body link and consider

- p_i is a generic point on the link
- The position of the **center of mass** is p_{l_i}

$$p_{l_i} = \frac{1}{m_i} \int_{V_i} p_i^* \rho dV_i$$

The kinetic energy of the link can be obtained with the following integral

$$T_i = \int_{V_i} \dot{p}_i^{*T} \dot{p}_i^*$$



Solving the integral yields

$$T_i = \frac{1}{2} m_i \dot{p}_{l_i}^T \dot{p}_{l_i} + \frac{1}{2} \omega_i^T I_i \omega_i$$

Kinetic energy of a link

Kinetic energy of a link

$$T_i = \frac{1}{2} m_i \dot{\mathbf{p}}_{l_i}^T \dot{\mathbf{p}}_{l_i} + \frac{1}{2} \boldsymbol{\omega}_i^T I_i \boldsymbol{\omega}_i$$

The expression of the **inertia tensor** I depends on the frame of reference

- In the **base frame**, the inertia tensor I_i **depends on the configuration** of the robot (joint variables)
- In a **frame rigidly attached to the link**, the inertia tensor is represented by a **constant matrix** ${}^i I_i$

$$I_i = R_i {}^i I_i R_i^T$$



From datasheet

Kinetic energy of a link

Kinetic energy of a link

$$T_i = \frac{1}{2} m_i \dot{\mathbf{p}}_{l_i}^T \dot{\mathbf{p}}_{l_i} + \frac{1}{2} \boldsymbol{\omega}_i^T I_i \boldsymbol{\omega}_i$$

Inertia matrix

$$I_i = R_i {}^i I_i R_i^T$$

Link velocity

$$\dot{\mathbf{p}}_i = J_P^{l_i}(\mathbf{q}) \dot{\mathbf{q}}_i$$

Angular velocity

$$\boldsymbol{\omega}_i = J_O^{l_i}(\mathbf{q}) \dot{\mathbf{q}}_i$$



$$T_i = \frac{1}{2} m_i \dot{\mathbf{q}}_i^T J_P^{l_i T} J_P^{l_i} \dot{\mathbf{q}}_i + \frac{1}{2} m_i \dot{\mathbf{q}}_i^T J_O^{l_i T} R_i {}^i I_i R_i^T J_O^{l_i} \dot{\mathbf{q}}_i$$

Kinetic energy of a link

The kinetic energy of all links can be combined to get the **total kinetic energy**

$$T_i = \frac{1}{2} m_i \dot{\mathbf{q}}_i^T J_P^{l_i T} J_P^{l_i} \dot{\mathbf{q}}_i + \frac{1}{2} m_i \dot{\mathbf{q}}_i^T J_O^{l_i T} R_i^i I_i R_i^T J_O^{l_i} \dot{\mathbf{q}}_i$$



$$T = \sum_{i=1}^n T_i = \frac{1}{2} \dot{\mathbf{q}}^T B(\mathbf{q}) \dot{\mathbf{q}} \quad \text{Total kinetic energy}$$

Where the **inertia matrix** of the manipulator is

$$B(\mathbf{q}) = \sum_{i=1}^n \left(m_i J_P^{l_i T} J_P^{l_i} + J_O^{l_i T} R_i^i I_i R_i^T J_O^{l_i} \right)$$

Potential energy of a link

The **potential energy** of a link is just related to the **gravitational force**

$$U_i = -m_i \mathbf{g}_0^T \mathbf{p}_{l_i}$$

Where $\mathbf{g}_0$ is the vector of gravitational acceleration in the base frame

The **total potential energy** of the manipulator is

$$U = \sum_{i=1}^n U_i = - \sum_{i=1}^n m_i \mathbf{g}_0^T \mathbf{p}_{l_i}$$

Dynamic model

Plugging the expressions of the total kinetic and potential energy of the robot into the lagrange equation, we obtain the canonical equation

$$B(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}$$

Inertia terms

$$B(\mathbf{q}) = \sum_{i=1}^n \left(m_i J_P^{l_i T} J_P^{l_i} + J_O^{l_i T} R_i^i I_i R_i^T J_O^{l_i} \right)$$

Gravitation terms

$$\mathbf{g}(\mathbf{q}) = \sum_{i=1}^n m_i \mathbf{g}_0^T J_{P_i}^{l_i}(\mathbf{q})$$

What about the **matrix C**? This matrix includes the **Coriolis and Centrifugal terms**, and its choice is not unique.

Computation of the matrix C

The matrix C is an $n \times n$ **matrix**, and in general it is **not symmetric**.

One possible definition is obtained using **Christoffel symbols** of the first type

$$C_{ij} = \sum_{k=1}^n C_{ijk} \dot{q}$$

Element of the matrix C at row i
and column j

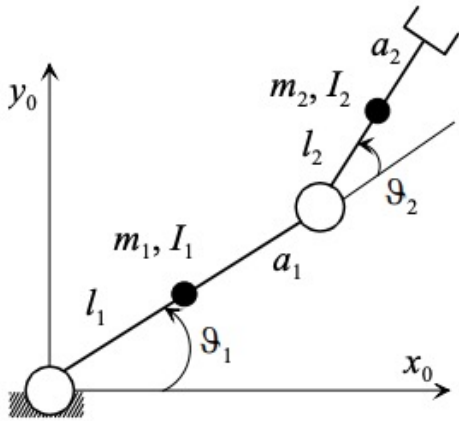
with

$$C_{ijk} = \frac{1}{2} \left(\frac{\partial B_{ij}}{\partial q_k} + \frac{\partial B_{ik}}{\partial q_j} - \frac{\partial B_{jk}}{\partial q_i} \right)$$

Christoffel symbol

This expression can be derived by expanding the velocity product terms in the Lagrange equation.

Two-link planar manipulator



- Mass of the two links: m_1 and m_2
- Length of the two links: a_1 and a_2
- Distance of the centers of mass from the joints: l_1 and l_2
- Moments of inertia around the axes passing through the centers of mass and parallel to z_0 : I_1 and I_2
- Generalized coordinates: $\mathbf{q} = [\theta_1, \theta_2]$

Position vectors (from the kinematics)

$$\mathbf{p}_{l_1} = \begin{bmatrix} l_1 c_1 \\ l_1 s_1 \\ 0 \end{bmatrix} \quad \mathbf{p}_{l_2} = \begin{bmatrix} a_1 c_1 + l_2 c_{12} \\ a_1 s_1 + l_2 s_{12} \\ 0 \end{bmatrix}$$

$$\mathbf{p}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{p}_1 = \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ 0 \end{bmatrix}$$

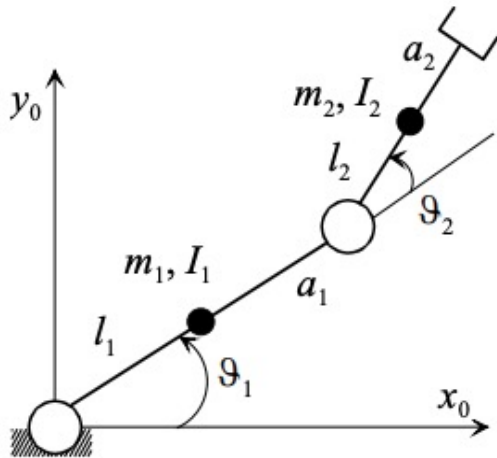
Axes

$$\mathbf{z}_0 = \mathbf{z}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Abbreviations

$$\begin{aligned} c_1 &= \cos(\theta_1) \\ c_{12} &= \cos(\theta_1 + \theta_2) \\ s_1 &= \sin(\theta_1) \\ s_{12} &= \sin(\theta_1 + \theta_2) \end{aligned}$$

Two-link planar manipulator



First, let's compute the **inertia matrix** $B(\mathbf{q}) = \sum_{i=1}^n \left(m_i J_P^{l_i T} J_P^{l_i} + J_O^{l_i T} \underbrace{R_i}_{\text{identity}} I_i R_i^T J_O^{l_i} \right)$

Note: the angular velocity vectors ω_1 and ω_2 are aligned with $\mathbf{z}_0$, so this rotation matrix is an identity

Link 1

$$J_P^{l_1} = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{p}_{l_1} - \mathbf{p}_0) & 0 \end{bmatrix} = \begin{bmatrix} -l_1 s_1 & 0 \\ l_1 c_1 & 0 \\ 0 & 0 \end{bmatrix} \quad J_O^{l_1} = \begin{bmatrix} \mathbf{z}_0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

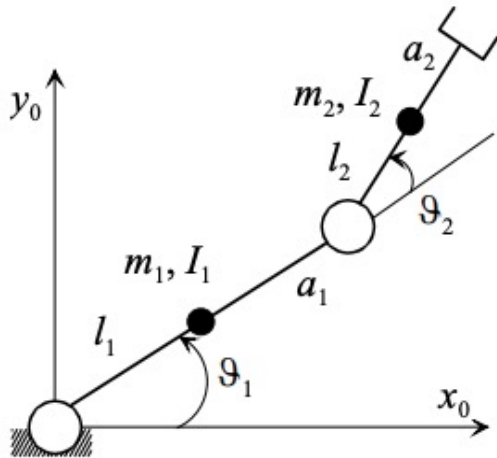
Link 2

$$J_P^{l_2} = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{p}_{l_2} - \mathbf{p}_0) & \mathbf{z}_0 \times (\mathbf{p}_{l_2} - \mathbf{p}_1) \end{bmatrix} = \begin{bmatrix} -a_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ a_1 c_1 + l_2 c_{12} & l_2 c_{12} \\ 0 & 0 \end{bmatrix} \quad J_O^{l_2} = \begin{bmatrix} \mathbf{z}_0 & \mathbf{z}_0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$



$$B(\mathbf{q}) = \begin{bmatrix} m_1 l_1^2 + I_1 + m_2(a_1^2 + l_2^2 + 2a_1 l_2 c_2) + I_2 & m_2(l_2^2 + a_1 l_2 c_2) + I_2 \\ m_2(l_2^2 + a_1 l_2 c_2) + I_2 & \underbrace{m_2 l_2^2 + I_2}_{\text{constant}} \end{bmatrix}$$

Two-link planar manipulator



Then, let's compute the **C matrix** $C_{ij} = \sum_{k=1}^n C_{ijk} \dot{q}$ with $C_{ijk} = \frac{1}{2} \left(\frac{\partial B_{ij}}{\partial q_k} + \frac{\partial B_{ik}}{\partial q_j} - \frac{\partial B_{jk}}{\partial q_i} \right)$

The only derivatives that are different from zero are:

$$\frac{\partial B_{11}}{\partial q_2} = -2m_2 a_1 l_2 s_2 = 2h$$

$$\frac{\partial B_{12}}{\partial q_2} = \frac{\partial B_{21}}{\partial q_2} = -m_2 a_1 l_2 s_2 = h$$

Therefore, in the computation of Christoffel Symbols we only need to account for these terms.

$$C_{111} = 0$$

$$C_{211} = -\frac{1}{2} \frac{\partial B_{11}}{\partial q_2} = h$$

$$C_{112} = C_{121} = \frac{1}{2} \frac{\partial B_{11}}{\partial q_2} = h$$

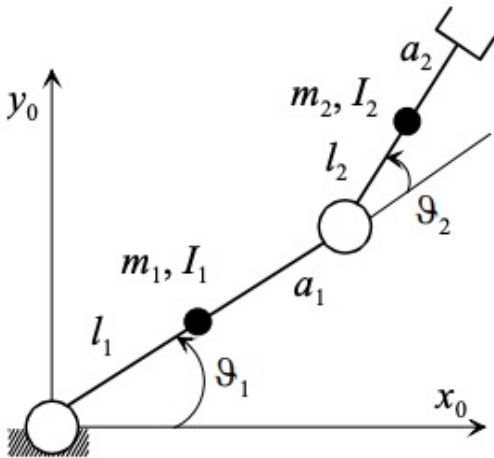
$$C_{212} = \frac{1}{2} \left(\frac{\partial B_{21}}{\partial q_2} - \frac{\partial B_{12}}{\partial q_2} \right) = 0$$

$$C_{122} = \frac{1}{2} \left(\frac{\partial B_{12}}{\partial q_2} + \frac{\partial B_{12}}{\partial q_2} \right) = h$$

$$C_{222} = 0$$

$$C(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} h \dot{q}_2 & h (\dot{q}_1 + \dot{q}_2) \\ -h \dot{q}_1 & 0 \end{bmatrix}$$

Two-link planar manipulator



Then, let's compute the **gravitational terms** $\mathbf{g}(\mathbf{q}) = \sum_{i=1}^n m_i \mathbf{g}_0^T J_{P_i}^{l_i}(\mathbf{q}) = \begin{bmatrix} g_1(\mathbf{q}) \\ g_2(\mathbf{q}) \end{bmatrix}$

The gravitational acceleration is $\mathbf{g}_0 = \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix}$

First term

$$g_1(\mathbf{q}) = -m_1 \mathbf{g}_0^T J_{P_1}^{l_1}(\mathbf{q}) - m_2 \mathbf{g}_0^T J_{P_1}^{l_2}(\mathbf{q}) = -m_1 \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix}^T \begin{bmatrix} -l_1 s_1 \\ l_1 c_1 \\ 0 \end{bmatrix} - m_2 \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix}^T \begin{bmatrix} -a_1 s_1 - l_2 s_{12} \\ a_1 c_1 + l_2 c_{12} \\ 0 \end{bmatrix} =$$

$$= (m_1 l_1 + m_2 a_1) g c_1 + m_2 g l_2 c_{12}$$

Second term

$$g_2(\mathbf{q}) = -m_1 \mathbf{g}_0^T J_{P_2}^{l_1}(\mathbf{q}) - m_2 \mathbf{g}_0^T J_{P_2}^{l_2}(\mathbf{q}) = -m_1 \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix}^T \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - m_2 \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix}^T \begin{bmatrix} -l_2 s_{12} \\ l_2 c_{12} \\ 0 \end{bmatrix} =$$

$$= m_2 g l_2 c_{12}$$

Two-link planar manipulator

In conclusion, the dynamic model of the two-link planar manipulator is

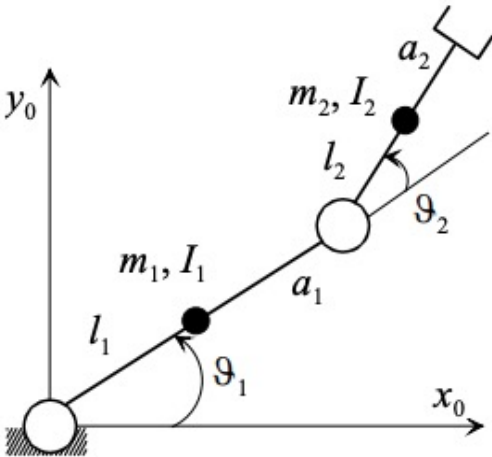
$$B(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}$$

with

$$B(\mathbf{q}) = \begin{bmatrix} m_1 l_1^2 + I_1 + m_2(a_1^2 + l_2^2 + 2a_1 l_2 c_2) + I_2 & m_2(l_2^2 + a_1 l_2 c_2) + I_2 \\ m_2(l_2^2 + a_1 l_2 c_2) + I_2 & m_2 l_2^2 + I_2 \end{bmatrix}$$

$$C(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -m_2 a_1 l_2 s_2 \dot{q}_2 & m_2 a_1 l_2 s_2 (\dot{q}_1 + \dot{q}_2) \\ m_2 a_1 l_2 s_2 \dot{q}_1 & 0 \end{bmatrix}$$

$$\mathbf{g}(\mathbf{q}) = \begin{bmatrix} (m_1 l_1 + m_2 a_1) g c_1 + m_2 g l_2 c_{12} \\ m_2 g l_2 c_{12} \end{bmatrix}$$



Friction and external forces

Besides the conservative gravitational forces there may be other forces acting on the manipulator

- **External force**: the effects of a force exerted on the mechanism at the end-effector can be accounted for by adding the term

$$J^T \mathbf{f} \quad \text{Right-hand side}$$

- **Viscous friction**: the viscous friction torque may be represented by a term in the form

$$-F_v \dot{\mathbf{q}} \quad \text{Left-hand side}$$

with F_v being a diagonal matrix of viscous coefficients

- **Static friction**: may be represented by a function

$$-\mathbf{f}_s(\mathbf{q}, \dot{\mathbf{q}}) \quad \text{Left-hand side}$$

Friction and external forces

The joint-space dynamic model for the manipulator with an external force, viscous and static friction is

$$B(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) - F_v\dot{\mathbf{q}} - \mathbf{f}_s(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau} + J^T \mathbf{f}$$



Inverse dynamics

$$\boldsymbol{\tau} = B(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) - F_v\dot{\mathbf{q}} - \mathbf{f}_s(\mathbf{q}, \dot{\mathbf{q}}) - J^T \mathbf{f}$$

Forward dynamics

$$\ddot{\mathbf{q}} = B(\mathbf{q})^{-1} (\boldsymbol{\tau} + J^T \mathbf{f}) - C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{g}(\mathbf{q}) + F_v\dot{\mathbf{q}} + \mathbf{f}_s(\mathbf{q}, \dot{\mathbf{q}})$$

Joint elasticity

- We may also model the **elasticity at the joints**. This means including in the Lagrangian two terms for the joint, one for the **motor kinetic energy** and one for the **elastic potential energy**.
- Let's denote the motor positions with θ
- The dynamic model becomes

$$B(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\mathbf{q} + \mathbf{g}(\mathbf{q}) + K(\mathbf{q} - \boldsymbol{\theta}) = \mathbf{0}$$

$2n$ generalized coordinates

No external force performing work on $\mathbf{q}$

$$B_m \ddot{\boldsymbol{\theta}} + K(\boldsymbol{\theta} - \mathbf{q}) = \boldsymbol{\tau}$$

Motor kinetic and potential terms

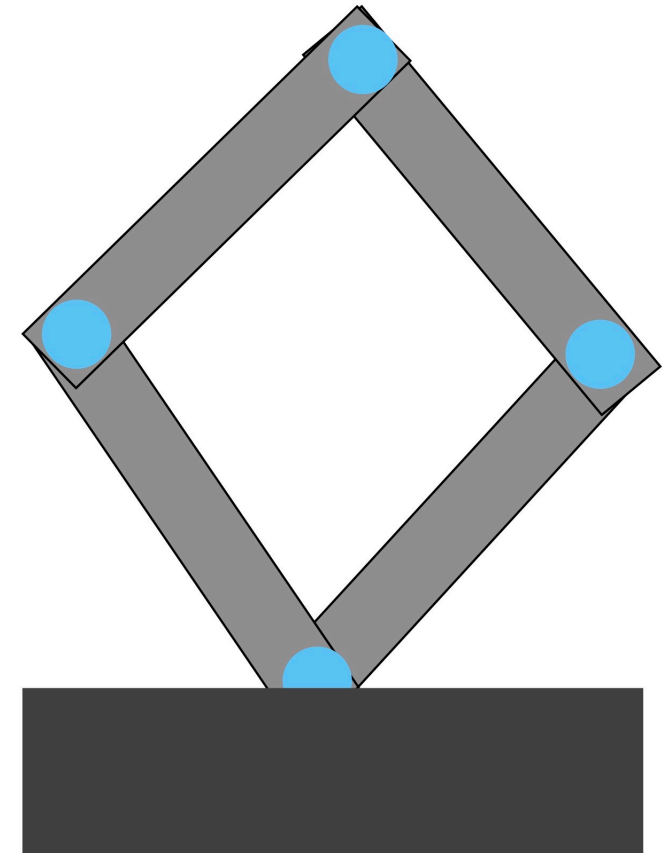
Kinematic Loops

In presence of **kinematic loops** ([parallel manipulator](#)), the **canonical equation in joint-space** takes the form

$$B(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau} + \boldsymbol{\tau}_a + \boldsymbol{\tau}_c$$

where

- $\boldsymbol{\tau}_a$ is the vector of [loop-closure active forces](#) (e.g., forces deriving from elastic, damping and actuation elements)
- $\boldsymbol{\tau}_c$ is the vector of [loop-closure constraint forces](#)



Kinematic Loops

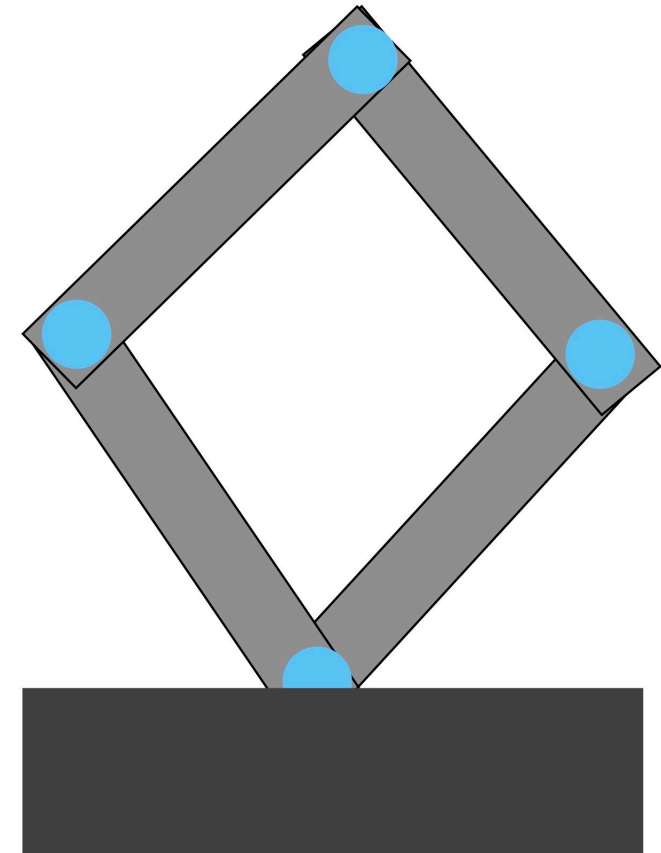
At the acceleration level, the loop-closure constraints can be represented in the form of a linear equation

$$L\ddot{\mathbf{q}} = \mathbf{1} \quad \text{with} \quad L \in \mathbb{R}^{n_c \times n}$$

Given this representation, the constraint forces can be written as

$$\boldsymbol{\tau}_c = L^T \boldsymbol{\lambda}$$

$\boldsymbol{\lambda}$ is a $(n_c \times 1)$ vector of **Lagrange multipliers**.



Kinematic Loops

Combining the canonical equation with the expression of the loop-constraints we get

$$\begin{pmatrix} B & L^T \\ L & 0 \end{pmatrix} \begin{pmatrix} \ddot{\mathbf{q}} \\ -\boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\tau} + \boldsymbol{\tau}_a - (C\dot{\mathbf{q}} + \boldsymbol{\tau}_g) \\ \mathbf{1} \end{pmatrix}$$

